

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 1st Term Examination-2020
Dying and Finishing I
LE 3107

Time: 1.5 Hours

Full Marks: 120

- N.B. i) Answer any **TWO** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|------|--|----|
| 1(a) | State the differences between bathochromic effect and hypsochromic effect. | 07 |
| 1(b) | Briefly describe, "Benzene is colorless, Nitrobenzene is pale yellow, and P-Nitroaniline is dark yellow". Write some limitations of Witt's theory. | 10 |
| 1(c) | Explain Quinonoid theory with limitations. Write the requisites of a true dye. | 13 |
| 2(a) | Define the followings: (i) Tint, (ii) Dye, (iii) Hue, (iv) Color index. Explain "Dispersol Blue Y-6B" according to color index. | 08 |
| 2(b) | Write short notes on: (i) Toxicity of dyes, (ii) H-acid, (iii) Banned amines and (iv) Resonance theory. | 12 |
| 2(c) | Sketch the operational sequence in dye and intermediate manufacturing. | 06 |
| 2(d) | State the effect of temperature on sulfonation process. | 04 |
| 3(a) | Why some azo dyes are banned? Prepare p-nitroaniline from acetanilide. | 08 |
| 3(b) | What are the differences between fluorescence and chemiluminescence? Explain Jabloniski diagram. | 12 |
| 3(c) | "Production of color through adsorption of radiation". Elucidate. | 10 |

SECTION-B

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|------|---|----|
| 4(a) | What are the key differences between crust leather and finished leather. | 10 |
| 4(b) | Which finishing coat is most responsible for the surface protection of leather? Explain why? | 08 |
| 4(c) | How will you achieve the desirable characteristics of leather prior to finishing operations. | 12 |
| 5(a) | Explain the factors related with pigment covering. Define coating. | 10 |
| 5(b) | Distinguish between prebase coat, base coat and intermediate coat. | 10 |
| 5(c) | How are the refraction index, shapes and sizes of the pigment particles important in leather coatings? | 10 |
| 6(a) | Graphically explain the film formation process involved in leather finishing. | 10 |
| 6(b) | Which adhesion mechanisms are more important for leather finishing and why? Justify your comments. | 10 |
| 6(c) | Provide some probable solutions to the following problems:
(i) Poor finish adhesion to leather and (ii) Inadequate pigment mix coverage. | 10 |

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 3rd Year 1st Term Examination-2020
Leather Products Engineering I
LE 3109

Time: 1.5 Hours

Full Marks: 120

N.B. i) Answer any **TWO** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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| 1(a) “What messages bear a particular dress pattern or fashion from man to man, community to community and overall economy to economy”- Explain with relevant example. | 08 |
| 1(b) What is clicking? State the qualities are required for a perfect clicker. | 08 |
| 1(c) Why interlocking system is important for cutting operation? Mention the cutting norms. | 14 |
| 2(a) Calculate the leather consumption of a complete Bi-fold wallet. | 10 |
| 2(b) Describe the following operations during fabrication of a wallet:
(i) Cradle Construction and (ii) Center Piece. | 10 |
| 2(c) Why allowance is important for leather goods manufacturing? | 06 |
| 2(d) Write down the concerning points for grading of leather garments. | 04 |
| 3(a) What are the causes of second wastage? How do you calculate leather wastage during cutting of materials? | 15 |
| 3(b) What are the differences among first wastage, second wastage and third wastage? | 06 |
| 3(c) Explain the characteristics of the following categories of fashion:
(i) Classic and (ii) Elegant. | 09 |

SECTION-B

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| 4(a) Justify 3D→2D→3D concept of pattern engineering for leather products construction with a neat sketch. | 12 |
| 4(b) Cite the parameters of garments grading. Why it is essential for the product development (PD) section of a leather garments industry? | 10 |
| 4(c) Classify design and leather goods with examples. How does grading differ from enlargement? | 08 |
| 5(a) Illustrate the Venn diagram to show the connection between design elements and principles. | 10 |
| 5(b) State the color specification parameters. Distinguish between tangible and intangible aspects of fashion. | 10 |

5(c)	Describe the trend for a fad to classical evolution of fashion based on time and adopter scale.	10
6(a)	State the suitable application of monochromatic and achromatic color schemes in leather outfit designing with neat sketches.	12
6(b)	List out the principal of Munsell colors. Interpret the Munsell color tree of 7PB 7/1.	10
6(c)	What is meant by the blueprint of product? Classify pattern according to garments.	08

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 1st Examination-2020
Footwear Engineering I
LE 3111

Time: 1.5 Hours

Full Marks: 120

N.B. i) Answer any two questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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| 1(a) Classify the bones of human foot. Briefly describe the tarsus bones of human foot with figures. | 10 |
| 1(b) Define articulation of anatomy. Illustrate the most critical articulation of human foot, along with its functions. | 05 |
| 1(c) Distinguish between pes planus and pes cavus, as well as the complications connected with each in wearing shoes. | 05 |
| 1(d) Write short notes on: (i) Achilles tendon (ii) Sustentaculum tali and (iii) Bursae | 10 |
| 2(a) Define supination and pronation of human foot. | 05 |
| 2(b) Illustrate the weight distribution pattern on average foot during standing and walking. | 07 |
| 2(c) "Correct shoes fitting to any foot requires heel-to- ball measurement"-Explain it. | 08 |
| 2(d) What is meant by Brannock device? Briefly state the foot measurement process using Brannock device. | 10 |
| 3(a) Define the girths of human foot including figure. | 12 |
| 3(b) Mention the names of international shoe sizing system. Describe paris point shoe sizing and fitting systems including the charts. | 12 |
| 3(c) Solve the following problems: (i) Convert ladies shoe size 75 to Paris point system (ii) Convert French point shoe size 26 to UK system (iii) Calculate the standard last length of English shoe size 9 for adults. | 06 |

SECTION-B

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|---|----|
| 4(a) Draw the pictures of Oxford shoe with basic parts. | 10 |
| 4(b) Make a complete list of shoe components with probable functions. | 12 |
| 4(c) Define the following adhesive terminologies:
(i) Green strength (ii) Reactivation life (iii) Open time and (iv) Peel strength | 08 |
| 5(a) For an ideal adhesive, higher cohesive strength, adhesive strength and drying rate are mandatory- justify the statement. | 15 |
| 5(b) To become a qualityful clicker of cutting department, which quality is mandatory from your point of view? | 15 |

- 6(a) Mention the name of different parts of splitting machine including their functions. 10
- 6(b) State the possible causes and remedy of the following problems during skiving 10
operation: (i) Improper material feeding (ii) Leather is getting teared
(iii) Intermittent material transportation
- 6(c) State the possible cause and remedy of the following problems during splitting 10
operation: (i) Unable to get required thickness (ii) Irregular material feeding (iii)
Unable to confirm uniform splitting

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 1st Examination-2020
Leather Manufacturing Technology IV
LE 3101

Time: 1.0 hour and 30 minute

Full Marks: 120

N.B. i) Answer any two questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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|------|---|----|
| 1(a) | What is fur tanning? State the principle of modern fur tanning. | 08 |
| 1(b) | List out the important characteristics of football leather. | 05 |
| 1(c) | How the drying operation influence on final leather quality? Briefly explain. | 07 |
| 1(d) | Describe the term in context of leather drying: | 10 |
| | (i) Equilibrium moisture content (ii) Falling rate period | |
| 2(a) | Write down the controlling factors and recipe formulation of deliming and bating operation for making of glove leather. | 08 |
| 2(b) | Why aging of pelt after pickling is recommended for glove leather production. Write in brief. | 06 |
| 2(c) | What should be your controlling techniques in liming operation for ideal Glace-kid leather production? Write with proper recipe formulation. | 08 |
| 2(d) | State the principle of manufacturing of upholstery leather. | 08 |
| 3(a) | Write about the choice of raw materials for the production of following leathers: | 10 |
| | (i) Clothing leather (ii) Suede leather (iii) Glove leather | |
| 3(b) | Write in details about the control of different manufacturing processes for the production of on ideal clothing leather. | 10 |
| 3(c) | Write about the control with proper recipe of the following tannery operation for suede leather manufacturing: (i) Chrome tanning and (ii) Dyeing | 10 |

SECTION-B

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|------|--|----|
| 4(a) | "Finishing is a mandatory process for leather manufacturing to upgrade the value of leather"-Justify with your comments. | 08 |
| 4(b) | Point out the effects of buffing and plating operation on leather finishing. | 08 |
| 4(c) | Write short notes on the term: (i) Conditioning (ii) Polishing | 08 |
| 4(d) | What are the problems that may occur due to deficient finishing? | 06 |
| 5(a) | Write the working principle of autospry machine and give the major controlling parameters. | 10 |
| 5(b) | Write down the instructions for glazing and embossing operations. | 08 |
| 5(c) | Write short notes on: (i) Filler (ii) Acrylic resin binder and (iii) Waxes and oils | 12 |
| 6(a) | Mention the classification of finishes according to finishing technique, materials and effects. | 09 |
| 6(b) | Give the objectives of the following coats: (i) Impregnation coat and (ii) Base coat. | 08 |

- 6(c) Write the objectives of top coat. What should be the material selection criteria for a standard top coat? 07
- 6(d) Compact binder is being used in leather finishing. Why you will choose a compact binder instead of a single binder? Discuss. 06